

RUTUJA BABAR

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Education

Master of Science in Computer Science

August 2024 – Present

Stony Brook University GPA: 3.45/4

Stony Brook, USA

Relevant Coursework: Distributed Systems, Analysis of Algorithms, Operating Systems, Adversarial Machine

Learning, Data Management

Bachelor of Computer Engineering

July 2020 – May 2024

University of Mumbai GPA: 9.42/10

Mumbai, India

Relevant Coursework: Data Structures, Data Warehousing and Mining, Computer Networks, Software Engineering

Technical Skills

- **Programming Languages:** Python, Java, Go, C++, JavaScript
- **Cloud Technologies:** AWS S3, EMR, Glue, EC2
- **Data Systems:** Apache Iceberg, Apache Spark, Apache Flink, REST Catalog, PyArrow, SQLite, SQL
- **Frameworks:** Node.js, React, TensorFlow, OpenCV
- **Developer Tools:** Git, Docker, LaTeX, IntelliJ, PyCharm, VS Code, macOS, Windows, Linux

Experience

Amazon Web Services

June 2025 – August 2025

SDE Intern - Open Data Analytics Engines

- Implemented equality delete read support in PyIceberg, enabling full Merge-on-Read functionality and compatibility with other Iceberg implementations.
- Developed a centralized DeleteFileIndex to optimize delete file organization and lookup efficiency in PyIceberg.
- Added validation logic to prevent partition field-schema name conflicts, ensuring consistency with Iceberg's Java and Rust implementations.
- Built comprehensive unit and integration tests for contributed features, improving code reliability and preventing regressions.
- Worked with team on design reviews and debugging to ensure features aligned with project goals and AWS standards.
- Contributed upstream to Apache Iceberg by implementing features and collaborating with the open-source community.

Stony Brook University

August 2024 – December 2024

Teaching Assistant - CSE 101: Computer Science Principles

- Conducted weekly lab sessions that covered computational thinking, Python programming, and algorithmic problem-solving.
- Assisted students with lab assignments and debugging, explaining core CS concepts.

The Sparks Foundation

February 2023 – March 2023

Web and Mobile Development Intern

- Developed a basic banking system website using HTML and JavaScript, and managed the database with MySQL, enhancing user transaction efficiency.
- Integrated a payment gateway into a donation website, enhancing the security and reliability of online transactions.

Projects

Modified Paxos for Intra-Shard Transactions | Go

September 2024 – October 2024

- Developed a consensus protocol for individual transactions without local logs, ensuring consistency in intra-shard operations.
- Engineered state synchronization and implemented prepare, promise, and accept phases.

Linear PBFT Implementation | Go

October 2024

- Designed a linearized PBFT protocol to ensure sequential transaction execution across distributed servers.
- Implemented leader election and transaction status tracking mechanisms for improved system transparency and fault tolerance.

Two-Phase Commit Protocol for Cross-Shard Transactions | Go, SQLite

November 2024 – December 2024

- Designed and implemented a two-phase commit protocol in Linear PBFT and Modified Paxos to handle transactions across distributed clusters, ensuring atomicity and isolation.
- Integrated inter-cluster communication mechanisms for synchronized and reliable transaction processing, enabling seamless execution of cross-shard operations.

Exercise Posture Recognition | React, Python, OpenCV, TensorFlow

August 2022 – April 2023

- Built a deep-learning-based web app to detect and correct user exercise posture in real time.
- Trained models with Keras and used MoveNet API for skeletal point detection.